

YI BA

Wuhan University of Technology
Automotive Engineering | **Email** | **Website**

EDUCATION

Wuhan University of Technology

B.Eng. in Automotive Engineering

GPA: 4.253/5.0 | Major Rank: 2/149 (Top 1.3%)

Wuhan, China

Sep 2024 – Expected Jun 2028

RESEARCH INTERESTS

Robot Learning; Autonomous Navigation; Robot Manipulation; Robot Perception

SUBMITTED MANUSCRIPT

Surface-Corridor-Based Intent-Aware Shared Control for Human-Guided Contact Motion on Curved Surfaces

Yi Ba, Chunhan Wang, Biwei Tang, Muye Pang, Jin Shi, and Jing Luo.

Submitted to *IEEE Transactions on Automation Science and Engineering*, 2026.

RESEARCH EXPERIENCE

Shanghai Jiao Tong University

Shanghai, China

Research Intern, supervised by Prof. Tong Qin

Jul 2026 – Present

DynaSCAN: Dynamic Collision-Aware Local Planning for Quadruped Robots in Crowded Environments

- Extended SCAN-Planner toward dynamic environments by integrating obstacle tracking and motion prediction with B-spline trajectory optimization for the Unitree Go2.
- Designed clearance-aware costs and multi-hypothesis trajectory initialization, with collision validation accounting for robot body geometry and turning dynamics.
- Implemented and evaluated the complete perception-to-planning pipeline in controlled dynamic-obstacle simulations, with broader system validation ongoing.

Wuhan University of Technology

Wuhan, China

Undergraduate Researcher, supervised by Prof. Jing Luo

Apr 2026 – Present

Shared Control for Curved-Surface Contact Manipulation

- Designed a trajectory-conditioned surface corridor from registered point clouds for geometry-consistent contact poses during human-guided motion.
- Developed confidence-weighted intent arbitration and integrated virtual fixtures with disturbance-observer (DOB)-enhanced normal admittance on a Sawyer robot.
- Validated the integrated framework through surface-frame comparison, controller ablation, and human-in-the-loop shared-control experiments.

SELECTED AWARDS AND HONORS

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| National First Prize, National Marine Vehicle Design and Production Competition | 2025 |
| National Second Prize, Global Campus Artificial Intelligence Algorithm Elite Competition | 2025 |
| Provincial First Prize, CIMC "Siemens Cup" China Intelligent Manufacturing Challenge | 2025 |

TECHNICAL SKILLS

Programming: Python, C++, MATLAB

Robotics: ROS 1/2, MoveIt, point-cloud processing

Simulation: MuJoCo, Isaac Gym, Gazebo

Tools: Git, Linux